N-ATLaS Integration Evidence

AtlasBridge: an OpenAI-compatible developer gateway and SDK for N-ATLaS

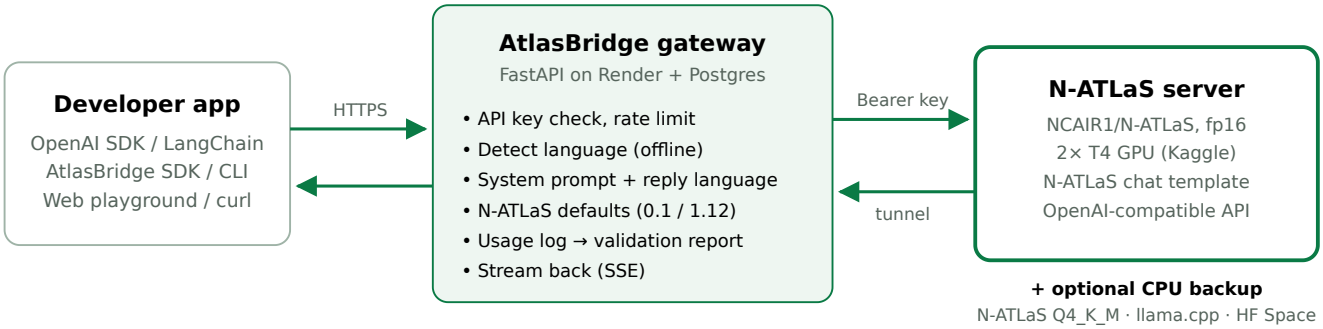
Challenge	National AI Innovation Challenge 2026, Innovation & Enterprise track
Problem statement	PS1: Developer Infrastructure
Applicant	Oyakhire Michael Kingsley, independent developer, Lagos
Repository	https://github.com/Kingsmichaei/atlasbridge
Live gateway	https://atlasbridge.onrender.com
Version	v2.1.0 (10 October 2026)

Summary. Every model call AtlasBridge makes goes to **N-ATLaS** (`NCAIR1/N-ATLaS` , the Llama-3-8B-based multilingual LLM published by Awarri Technologies and FMCIDE). No other foundation model is used or wrapped. AtlasBridge loads the official weights and applies N-ATLaS's own chat template and the generation settings from its model card. It then exposes the model through an OpenAI-compatible API, a Python SDK and a web playground, so developers can build on N-ATLaS without owning a GPU.

1. Which part of N-ATLaS is integrated

N-ATLaS component	Used?	How
N-ATLaS LLM (`NCAIR1/N-ATLaS`)	**Yes, core**	Official weights downloaded from Hugging Face after accepting the access conditions. Served in fp16 across 2× NVIDIA T4 GPUs. Handles all chat, translation and model-based language identification.
N-ATLaS chat template	**Yes**	Prompts are built with the model's own `tokenizer.apply_chat_template(..., date_string=...)` , exactly as the model card shows. We don't use a hand-written prompt format.
Recommended generation settings	**Yes**	Defaults follow the model card: `temperature=0.1` , `repetition_penalty=1.12` , stop on `<|eot_id|>` .
N-ATLaS Q4_K_M GGUF (`tosinamuda/N-ATLaS-GGUF` , a quantisation of `NCAIR1/N-ATLaS`)	**Built, optional**	Optional CPU backup served by llama.cpp. The gateway's automatic failover to it is built and tested, and it can be enabled on any server with about 8 GB of RAM. It isn't deployed for this submission because free hosting for it is no longer available.
Supported languages	**Yes**	The API is scoped to the four languages N-ATLaS supports: English, Hausa, Igbo and Yoruba.
N-ATLaS ASR models	No	Not needed for PS1. The gateway's design (one upstream per capability) allows an ASR endpoint to be added later.

2. Architecture and request flow



Every chat, translate and model-based detect request ends at N-ATLaS. There is no other model in the path.

3. Integration in the code

The excerpts below are from the repository. File paths are relative to the repo root.

3.1 Loading the official N-ATLaS weights

`serving/fallback_server.py`

```
MODEL_ID = "NCAIR1/N-ATLaS"
tok = AutoTokenizer.from_pretrained(MODEL_ID, token=HF_TOKEN)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID, torch_dtype=torch.float16, device_map="auto", token=HF_TOKEN)
# stop on Llama-3 end-of-turn as well as EOS
EOS_IDS = [tok.eos_token_id, tok.convert_tokens_to_ids("<|eot_id|>")]
```

3.2 N-ATLaS's own chat template and model-card generation settings

`serving/fallback_server.py`

```
text = tok.apply_chat_template(
    messages, add_generation_prompt=True, tokenize=False,
    date_string=datetime.now().strftime("%d %b %Y"))      # as in the model card
inputs = tok(text, return_tensors="pt", add_special_tokens=False).to("cuda:0")
kwargs = dict(max_new_tokens=max_tokens,
              repetition_penalty=float(body.get("repetition_penalty", 1.12)),
              temperature=float(body.get("temperature", 0.1)), ...)
```

3.3 Gateway: every request is forwarded to N-ATLaS

`gateway/app/upstream.py`

```
def build_payload(body: dict, stream: bool) -> dict:
    """Forward only the parameters N-ATLaS understands, with N-ATLaS-friendly defaults."""
    return {"model": settings.upstream_model,            # "n-atlas"
            "messages": body["messages"], "stream": stream,
            "temperature": body.get("temperature", 0.1),
            "max_tokens": body.get("max_tokens") or 512,
            "repetition_penalty": body.get("repetition_penalty", 1.12)}
```

3.4 Resilience: automatic failover between N-ATLaS servers

gateway/app/upstream.py

```
def backends() -> list[Backend]:
    out = []
    if settings.upstream_url:   out.append(Backend("gpu", ...))          # N-ATLaS fp16, Kaggle
    if settings.fallback_url:   out.append(Backend("cpu-backup", ...))   # N-ATLaS Q4_K_M,
llama.cpp
    return out
# healthy backends first (health cached 20 s); a failed backend is marked down and the next is
tried
```

Each response reports which N-ATLaS backend served it (`x_atlasbridge.backend`), and the usage log records it. The failover is covered by automated tests. In this submission only the GPU backend is deployed.

3.5 Making N-ATLaS reliable in multilingual apps

In testing, N-ATLaS sometimes replied in the wrong Nigerian language when left to infer it, for example answering an Igbo question in Hausa after an earlier Hausa turn. AtlasBridge detects the language of each message offline and names the reply language in the system prompt sent to N-ATLaS:

gateway/app/main.py (_prepare_messages)

```
instruction = (f"The user's latest message is in {name}. Reply only in {name}, even if earlier "
               f"messages used another language, unless the user explicitly asks for a different
language.")
```

The offline detector scores **29/30 (97%)** on a held-out set of everyday phrases in Hausa, Igbo and Yoruba, typed with or without diacritics. Developers can also set the reply language explicitly with `"language": "ha" | "ig" | "yo" | "en"`.

3.6 Translation and language ID as N-ATLaS tasks

gateway/app/main.py

```
system = (f"You are a professional translator. Translate the user's {src_name} text into
{tgt_name}. "
          f"Output only the {tgt_name} translation, with no explanations or quotes.")
text, usage = await _ask(system, body.text, ...)        # → N-ATLaS
```

4. Live evidence

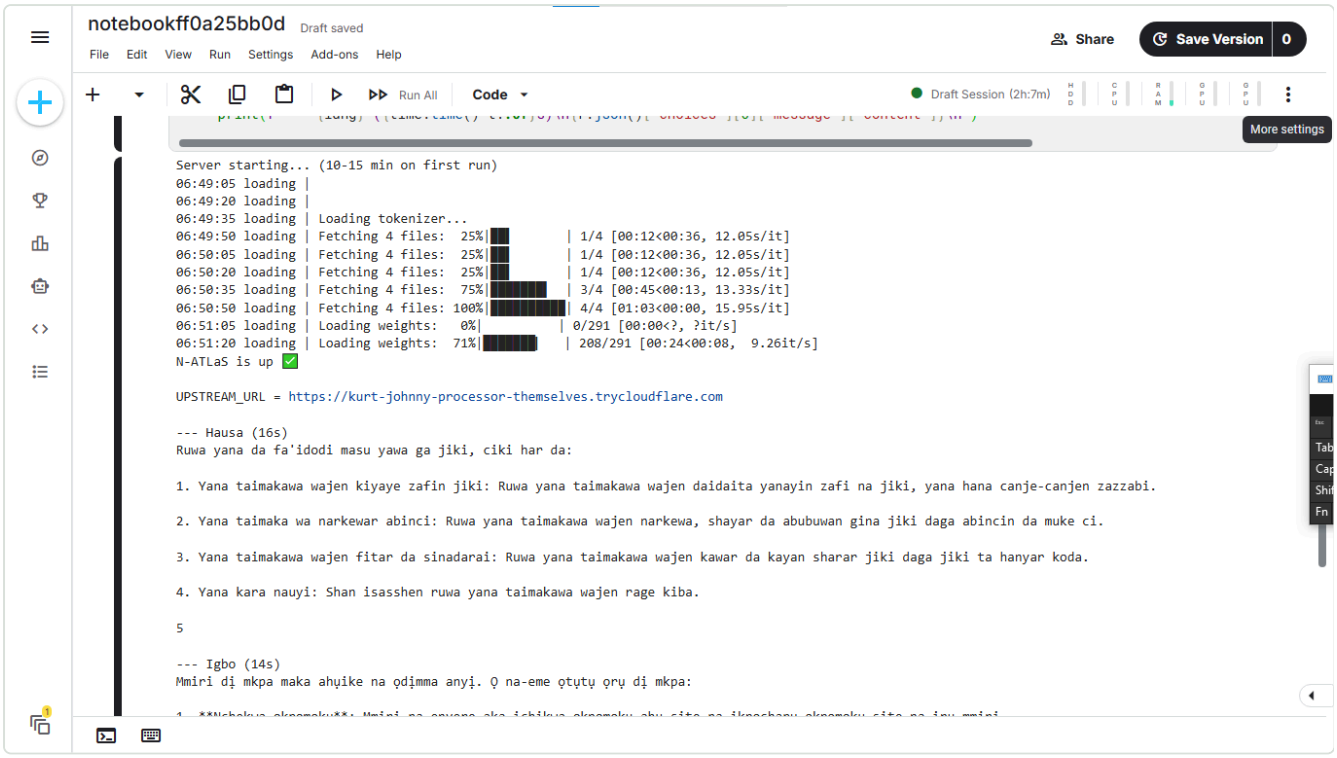


Figure 1. The serving notebook on Kaggle (2× T4), 10 October 2026. The four official N-ATLaS weight shards download and load (291 tensors), the server reports "N-ATLaS is up", the public tunnel URL is issued, and N-ATLaS answers the smoke-test prompts. Hausa: 16 s; the Igbo reply begins below it.

The smoke test sends the same question in all four languages. Excerpt of N-ATLaS's Hausa reply from Figure 1:

```
Prompt (Hausa):  Menene amfanin ruwa ga jikin dan Adam?   ("What are the benefits of water to the human body?")
N-ATLaS:         Ruwa yana da fa'idodi masu yawa ga jiki, ciki har da:
                 1. Yana taimakawa wajen kiyaye zafin jiki ...
                 2. Yana taimaka wa narkewar abinci ...
```

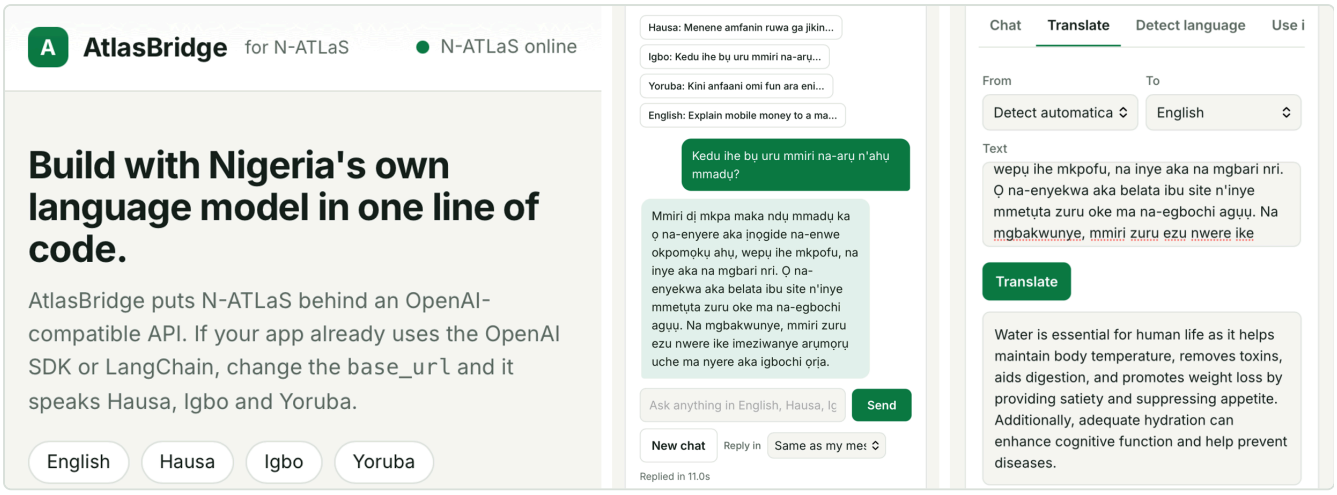


Figure 2. The live playground at https://atlasbridge.onrender.com, 10 October 2026. Left: status "N-ATLaS online". Middle: an Igbo question ("What is the use of water in the human body?") answered by N-ATLaS in Igbo in 11.0 s. Right: that Igbo answer translated to English through `/v1/translate`, which is also served by N-ATLaS.

5. How the panel can verify the integration

While the N-ATLaS server is running, the panel can check the integration without our help:

1. **Gateway health:** `GET https://atlasbridge.onrender.com/health` returns `"upstream": {"status": "ok", "backend": "gpu", ...}` while the N-ATLaS server is running. Every response carries the header `X-Powered-By: N-ATLaS (Awarri Technologies / FMCIDE)`.
2. **Model listing:** `GET /v1/models` returns the single model `n-atlas` with languages `en, ha, ig, yo`.
3. **Try it:** open https://atlasbridge.onrender.com, click *Join the beta* (invite code supplied in the application), and chat in Hausa, Igbo or Yoruba.
4. **Run it yourself:** the notebook `serving/atlasbridge_serve_kaggle.ipynb` reproduces the model server from the official weights on any Kaggle account with access to `NCAIR1/N-ATLaS`.
5. **Read the code:** the paths in section 3 contain the integration logic. `gateway/tests/` holds 23 automated tests, including the official OpenAI Python SDK running unchanged against the gateway.

Availability. The gateway, playground, documentation and repository are online at all times. The N-ATLaS model runs on a free Kaggle GPU (sessions of up to 12 hours), so it isn't always on. During the verification window (15–17 October) we will keep it running, and it can also be started on request within about 10 minutes: contact kingsmichael.official@gmail.com. While it's off, the playground shows "Model offline" and the API returns a clear 502/503 error.

6. Licence and attribution compliance

- Model access was obtained by accepting the N-ATLaS conditions on Hugging Face. The weights are not redistributed. They are downloaded from the official repository at runtime.
- The required attribution appears in the playground footer, the API description, the README, the SDK, and an HTTP header on every response.

- AtlasBridge is a beta under the licence's 1,000-active-user cap. Commercial scale-up would go through Awarri's commercial licence.
- Use is limited to the licence's permitted purposes (developer tooling, education, civic and community innovation). Beta testers are asked not to submit personal data.

N-ATLaS is an initiative of the Federal Ministry of Communications, Innovation and Digital Economy, and powered by Awarri Technologies. AtlasBridge is an independent developer tool built on N-ATLaS.